

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{DD} = 10\text{ V}$; $R_{B1} = 100\text{ k}\Omega$; $R_{B2} = 150\text{ k}\Omega$; $R_E = 1\text{ k}\Omega$; $R_C = 2.7\text{ k}\Omega$; $R_S = 0.68\text{ k}\Omega$; $R_F = 10\text{ k}\Omega$; $R_g = 1\text{ k}\Omega$; $C_g = 100\text{ nF}$.

Q_1 (pnp BJT): $V_{BE} = -0.7\text{ V}$; $\beta_F = \beta_0 = 150$; $r_0 = +\infty$;

M_2 (e-NMOS): $V_{tN} = 3\text{ V}$; $I_{DSSN} = 0.9\text{ mA}$; $r_0 = +\infty$;

$V_T = 26\text{ mV}$

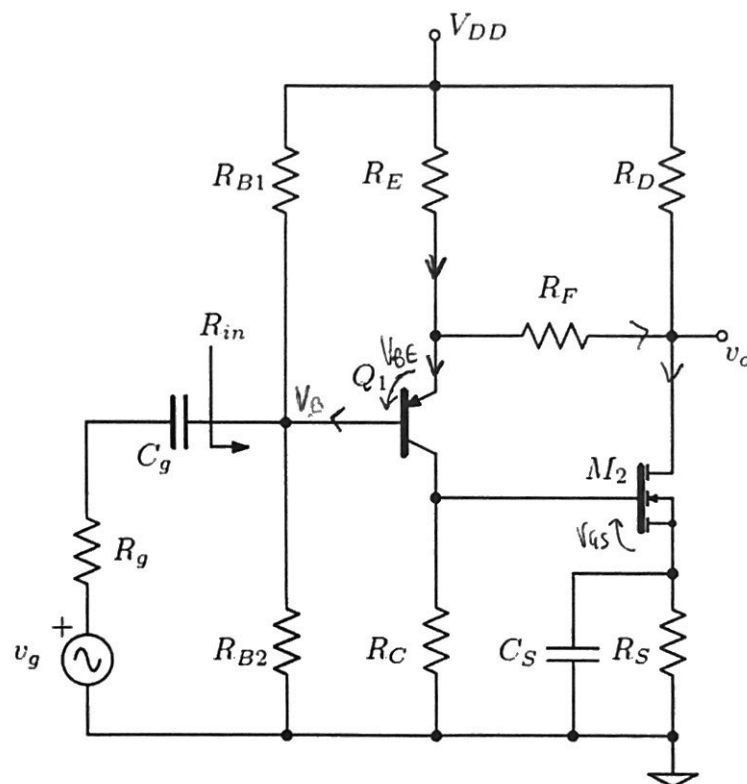


Fig. 1 – A case of two-stage feedback amplifier.

Considering *all capacitors* to be equivalent to *open circuits*, determine:

1. the value of resistance R_D that makes $V_O = V_{DD}/2$ in the steady state;
2. the operating points (V_{CE} , I_C) and (V_{DS} , I_D) of Q_1 and M_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), both C_g and C_S can be considered equivalent to *short circuits* determine also:

3. the voltage gain $A_v = v_o/v_g$ of the amplifier;
4. the input resistance R_{in} indicated in the figure.

Finally, considering only the effect of capacitor C_g (assumed to be dominant), find:

5. the low frequency bandwidth limit f_L of the amplifier.

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 12 \text{ k}\Omega$; $C_1 = 10 \text{ nF}$.

OpAmp differential gain: $A_{d0} = 10^5 \text{ [V/V]}$, $\omega_1 = 100 \text{ rad/s}$.

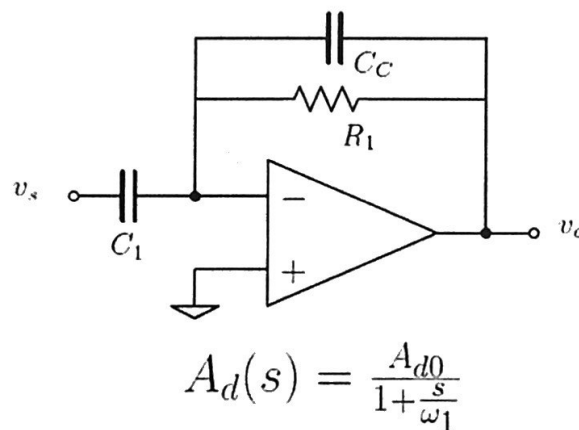


Fig. 2 – Compensated OpAmp in the inverting configuration.

Considering, initially, C_c **equivalent to an open circuit**, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the crossover frequency and phase margin of the amplifier;
3. the block diagram of the amplifier *according to feedback theory*.

Then, considering the effect of C_c , determine:

4. the value of C_c needed to get a 45° phase margin;
5. the Bode plot of the compensated amplifier gain.